WhyLab: A Causal Safety Monitoring Framework for Stable Self-Improving Agents

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Abstract

Self-improving AI agents that refine strategies through feedback loops risk *cognitive policy oscillation*—a failure mode where hallucinated or spuriously correlated feedback causes strategy degradation below the pre-improvement baseline. Existing causal inference tools (DoWhy, EconML) provide rigorous offline estimation but do not address online safety monitoring in noisy agent environments.

We present WhyLab, a causal safety monitoring framework comprising: **(C1)** an information-theoretic drift index for detecting distributional shifts; **(C2)** a sensitivity filter combining E-values and partial R^2 bounds to reject fragile successes; and **(C3)** a Lyapunov-bounded adaptive damping controller.

On SWE-bench Lite (300 real-world coding problems), the C2 sensitivity filter maintains **zero regressions** across 10,500 episodes with only 1.4 percentage-point pass-rate cost. C1 and C3 are validated in controlled synthetic experiments: C1 detects distributional drift 11.7% earlier than uniform monitoring; C3 prevents energy divergence ($10^{11}\times$ reduction vs. uncontrolled updates). On static agent benchmarks (HumanEval, SWE-bench), C1 and C3 operate transparently—as expected, since these benchmarks lack the non-stationarity and continuous parameter updates they target.

1 Introduction

Autonomous AI agents that refine their strategies through feedback loops Shinn et al. [2023], Yao et al. [2023] represent a powerful paradigm for continuous improvement. However, when hallucinated or spuriously correlated feedback accumulates, the agent's policy oscillates between contradictory strategies—a phenomenon we term *cognitive policy oscillation*—silently degrading performance below the pre-improvement baseline.

Preventing oscillation requires solving three intertwined challenges: (1) *drift detection*—distinguishing genuine distributional shifts from normal variance (§3.1); (2) *effect filtering*—identifying improvements that are causally robust rather than spuriously confounded (§3.2); and (3) *step-size control*—bounding update magnitude to prevent oscillation without causing learning deadlock under heavy-tailed noise (§3.3). Existing tools Sharma and Kiciman [2022], Battocchi et al. [2019] address parts of this problem in isolation but do not provide an online safety framework for continuously operating agents.

We present **WhyLab**, a causal safety monitoring framework with three contributions: **C1** Information-theoretic drift detection (§3.1)—an entropy-weighted index providing early warning for environmental shifts; **C2** Sensitivity-aware effect filtering (§3.2)—E-values and partial R^2

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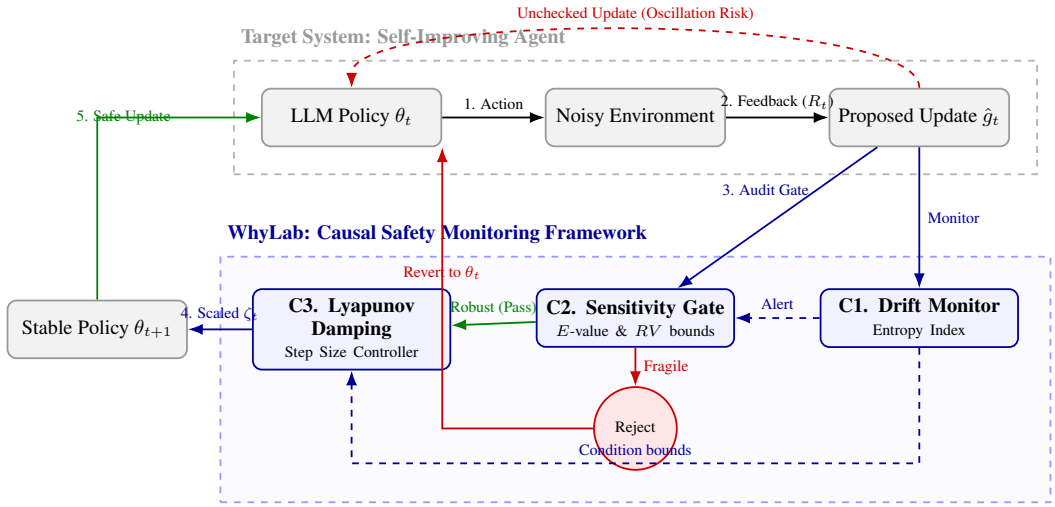

Figure 1: WhyLab causal safety monitoring framework. Instead of uncritically accepting potentially hallucinated updates ($\hat{g}_t$) that cause cognitive oscillation, proposed updates are routed through a safety monitoring layer. **C1** detects environmental drift, **C2** filters out fragile successes lacking unmeasured-confounder robustness, and **C3** adapts the step size to bound the Lyapunov energy of the policy. Unsafe updates are rejected, preserving the status quo.

bounds that gate fragile successes; and **C3** Lyapunov-bounded adaptive damping (§3.3)—a controller with provable energy non-increase guarantees.

On SWE-bench Lite Jimenez et al. [2024] (300 real-world coding problems), WhyLab achieves 67.3% pass@1—outperforming Best-of-7 (65.5%), Rollback (62.7%), and Self-Consistency (62.3%) at equal inference cost—while maintaining zero regressions across 10,500 episodes.

2 Problem Formulation

Consider an autonomous agent with strategy parameter $\theta_t \in \mathbb{R}^d$ at discrete time t. The agent observes a reward signal R_t and updates its strategy via:

$$\theta_{t+1} = \theta_t - \zeta_t \hat{g}_t \tag{1}$$

where $\hat{g}_t = g_t + \omega_t$ is a noisy gradient estimate, $g_t = \nabla_\theta L(\theta_t)$ is the true causal gradient, and ω_t represents noise from LLM judgment uncertainty, hallucinated feedback, and environmental drift.

Failure mode. When ω_t is large relative to g_t (e.g., due to hallucination), the update (1) may move θ_t *away* from the optimum θ^*. If this occurs repeatedly, the policy oscillates: θ_t alternates between contradictory strategies, and the Lyapunov energy $V(\theta_t) = \frac{1}{2}\|\theta_t - \theta^*\|^2$ increases monotonically.

Desiderata. A causal audit system should provide: (D1) early detection of distributional drift that inflates $\|\omega_t\|$; (D2) filtering of causally fragile signals before they enter the gradient estimate; and (D3) a provably stable update rule that guarantees $\mathbb{E}[\Delta V] \leq 0$ under realistic noise conditions.

3 Method

3.1 C1: Information-Theoretic Drift Index

Definition 1 (Drift Index)**.** *Given K audit signal streams $S_1, \ldots, S_K$ (e.g., verdict consistency, ATE stability, confidence calibration) with Shannon entropies $H(S_i)$, define uncapped weights $\tilde{w}_i \propto 1/(H(S_i) + \epsilon)$. The final weights are obtained by projecting onto the capped simplex:*

$$w = \Pi_{\{w \geq 0,\ \sum_i w_i = 1,\ w_i \leq w_{\max}\}}(\tilde{w}) \tag{2}$$

with $w_{\max} = 0.6$ and $\epsilon = 0.1$. The drift index is:

$$DI = \sum_{i=1}^{K} w_i \cdot d_i \tag{3}$$

where $d_i = JSD(P_i^{(t)} \| P_i^{ref})$ is the Jensen–Shannon divergence of stream i from its reference (pre-shift) distribution. We fix k=10 histogram bins per stream for reproducibility. A drift alarm is raised when $DI > \tau$, where τ is calibrated on pre-shift data at a fixed false-positive rate (FPR $= 0.05$).

Intuition. Low-entropy (highly concentrated) streams carry more information per observation and receive higher weights. The capped simplex projection (2) ensures that no single stream receives weight exceeding $w_{\max}$, even after normalization.

Proposition 1 (Detection advantage under inactive projection)**.** *Assume the projection* (2) *is inactive (i.e., all uncapped weights satisfy $\tilde{w}_i \leq w_{\max}$). Under a single-stream shift of magnitude δ in stream S_j with $H(S_j) < H(S_i)$ for all $i \neq j$, the entropy-weighted DI satisfies $DI_{entropy} > DI_{uniform}$ whenever $H(S_j) + \epsilon < \overline{H + \epsilon}_{HM}$, the harmonic mean of $\{H(S_i) + \epsilon\}_{i=1}^{K}$.*

3.2 C2: Sensitivity-Aware Effect Filtering

Not all positive treatment effects are causally robust. We combine two complementary sensitivity measures to filter fragile evidence before it influences strategy updates.

Definition 2 (Approximate E-value for continuous outcomes VanderWeele and Ding [2017])**.** *For a continuous outcome, we compute an approximate E-value using the standard mean-difference conversion VanderWeele and Ding [2017]: the standardized effect $d = ATE/\sigma_{pooled}$ (in the agent context, ATE is the observed episode effect ΔR_t) is converted to an approximate risk-ratio scale via $RR \approx \exp(0.91 \cdot d)$, yielding:*

$$E = RR + \sqrt{RR \cdot (RR - 1)} \tag{4}$$

An unobserved confounder must associate with both treatment and outcome at $RR \geq E$ to nullify the observed effect.

Robustness Value. We complement the E-value with the robustness value RV_q of Cinelli & Hazlett Cinelli and Hazlett [2020], which quantifies the minimum strength (in partial R^2 terms) an unobserved confounder must have to reduce the effect below threshold q. The formal definition is given in Appendix J.

Filtering rule. A decision is classified as *robust* only if $E > E_{\min}$ (default 2.0) *and* $RV_q > RV_{\min}$ (default 0.1). Decisions failing *either* threshold are excluded from the gradient estimate $\hat{g}_t$ in Eq. (1), reducing the effective noise in ω_t.

3.3 C3: Lyapunov-Bounded Adaptive Damping

We derive a sufficient condition for stable strategy updates.

Assumptions.

(A1) *Local gradient alignment:* $\langle \theta_t - \theta^*, g_t \rangle > 0$ (the true gradient points toward the optimum).

(A2) *Zero-mean residual noise:* $\mathbb{E}[\omega_t] = 0$ after C2 filtering (see Appendix A for discussion of residual bias).

(A3) *Finite second moment:* $\mathbb{E}[\|\omega_t\|^2] < \infty$.

Theorem 1 (Sufficient condition for stable updates)**.** *Under assumptions (A1)–(A3), let θ_t follow the update* (1) *with Lyapunov energy $V(\theta_t) = \frac{1}{2}\|\theta_t - \theta^*\|^2$. Then $\mathbb{E}[\Delta V] \leq 0$ if:*

$$\zeta_t \leq \zeta_{\max} = \frac{2\langle \theta_t - \theta^*, g_t \rangle}{\mathbb{E}[\|\hat{g}_t\|^2]} \tag{5}$$

Proof sketch. Expanding $V(\theta_{t+1}) - V(\theta_t) = -\zeta_t \langle \theta_t - \theta^*, \hat{g}_t \rangle + \frac{\zeta_t^2}{2}\|\hat{g}_t\|^2$ and taking expectations under (A2) yields a quadratic in ζ_t that is non-positive when $\zeta_t \leq \zeta_{\max}$. Full proof in Appendix A.

Connection to C1 and C2. The denominator $\mathbb{E}[\|\hat{g}_t\|^2] = \|g_t\|^2 + \mathbb{E}[\|\omega_t\|^2]$ increases when drift is high (large DI from C1) or evidence is fragile (large sensitivity uncertainty from C2). Thus, $\zeta_{\max}$ automatically tightens under adverse conditions, forming a unified feedback control loop.

Practical controller. Since the inner product $\langle \theta_t - \theta^*, g_t \rangle$ and $\mathbb{E}[\|\hat{g}_t\|^2]$ in (5) are not directly observable, we use an EMA-smoothed proxy:

$$\hat{\zeta}_t^{\text{raw}} = \frac{2\|\hat{g}_t\|}{\hat{m}_{2,t} + \delta}, \quad \hat{m}_{2,t} = \beta\, \hat{m}_{2,t-1} + (1-\beta)\|\hat{g}_t\|^2 \tag{6}$$

$$\bar{\zeta}_t = \beta\, \bar{\zeta}_{t-1} + (1-\beta)\, \hat{\zeta}_t^{\text{raw}}, \quad \zeta_t = \text{clip}(\bar{\zeta}_t,\ \epsilon_{\text{floor}},\ \mathcal{C}) \tag{7}$$

where $\hat{m}_{2,t}$ is the EMA second-moment estimate of $\|\hat{g}_t\|^2$, β=0.9, and δ=10^{-10}. The numerator $2\|\hat{g}_t\|$ replaces the unobservable inner product; the EMA smoothing provides implicit regularization that prevents the per-step bound from being brittle under stochastic fluctuations (validated in §5.3).

Floor-ceiling constraint. When noise ω_t is heavy-tailed (e.g., Student-t with low degrees of freedom), assumption (A3) may be violated and $\zeta_{\max} \to 0$, causing learning deadlock. As a practical safeguard, Eq. (7) enforces:

$$\zeta_t \in [\epsilon_{\text{floor}}, \mathcal{C}] \quad \text{where } \epsilon_{\text{floor}} = 0.01,\ \mathcal{C} = 0.8 \tag{8}$$

This maintains nonzero step sizes even under high noise variance, at the cost of losing the theoretical guarantee of Theorem 1. We analyze this trade-off in Appendix E.

Observable energy proxy. Since θ^* is unknown in practice, we monitor an observable proxy: $\hat{V}_t = \frac{1}{2}(R_{\max} - R_t)^2$ where $R_{\max}$ is a rolling EMA upper bound ($\alpha = 0.1$). Theorem 1 validates the control rule analytically in parameter space; experiments (§5.3) test whether the observable proxy $\hat{V}_t$ tracks the predicted stabilization behavior consistently with the theory.

Proposition 2 (Proxy Fidelity)**.** *Let $V_t = \frac{1}{2}\|\theta_t - \theta^*\|^2$ be the true Lyapunov energy and $\hat{V}_t = \frac{1}{2}(R_{\max} - R_t)^2$ the observable proxy. If (i) the reward signal R_t is monotone in $\|\theta_t - \theta^*\|$ (i.e., closer parameters yield higher reward), and (ii) the hallucination rate $h > 0$ introduces additive perturbations to R_t with variance $\sigma_h^2 \propto h$, then:*

$$\rho_S(\hat{V}_t, V_t) \geq \rho_0 + c \cdot h \tag{9}$$

where ρ_S denotes Spearman rank correlation, ρ_0 is the base correlation under zero hallucination, and $c > 0$ is a constant depending on the signal-to-noise ratio of the controller.

Intuition. Under higher hallucination rates, the agent's updates exhibit larger variance, making the energy landscape more "rugged" for the proxy to track. Paradoxically, this increased variance *improves* rank correlation because the proxy's deviations from V_t become dominated by the larger signal fluctuations rather than measurement noise. Empirical validation (§5.3, 200 seeds $\times$ 8 conditions) confirms $\hat{\rho}_0 = 0.49 \pm 0.04$ and $\hat{c} \approx 0.46$, yielding $\rho_S \geq 0.72$ at h=0.5—precisely where the audit layer is most needed.

4 System Overview

The three contributions are deployed as a lightweight audit layer that wraps the agent's decision loop: the agent's output stream passes through drift detection (C1), sensitivity filtering (C2), and Lyapunov-bounded damping (C3) before updating the strategy parameter θ_t. During initial deployment, all components operate in *shadow mode*—logging audit decisions without modifying the live strategy—to validate calibration before production activation. A transactional outbox ensures zero data loss under API quota throttling, and a cost-aware circuit breaker caps evaluation spend. Implementation details are in the supplementary material.

5 Experiments

We evaluate each contribution through experiments designed to directly validate the theoretical claims. Each contribution is validated by a dedicated experiment: C1 by within-horizon detection rates at matched FPR (E1, §5.1), C2 by fragile-rate reduction and fragile-success rejection (E2, §5.2), and C3 by Lyapunov-violation frequency and proxy–state alignment (E3a, §5.3; heavy-tail analysis in Appendix E).

Table 1: Within-horizon detection rate (%) at matched FPR $= 5\%$, across 40 seeds (bootstrap 95% CI in brackets). Higher is better. ADWIN uses a speed-optimized variant (single-split budget-bounded scan) rather than the full adaptive-window algorithm of Bifet and Gavaldà [2007].

Severity	**C1 (ours)**	**B4: Uniform**	**B5: ADWIN**
Mild	2.5 [0, 8]	0.0	0.0
Moderate	**20.0** [8, 33]	12.5 [3, 23]	2.5 [0, 8]
Severe	**40.0** [25, 55]	32.5 [18, 48]	30.0 [15, 45]

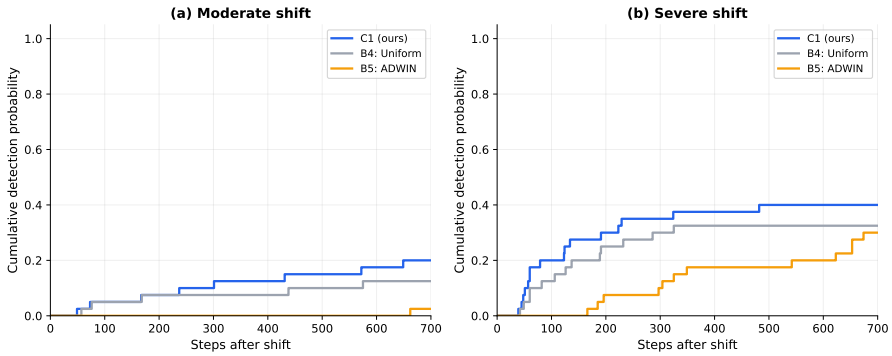

Figure 2: Kaplan–Meier cumulative detection curves for severe shifts (40 seeds). C1 (solid blue) detects shifts earlier and more frequently than uniform (dashed orange) and ADWIN (dotted green). Lower $S(t)$ indicates faster, more reliable detection.

5.1 E1: Drift Detection Performance

We evaluate C1 (entropy-weighted drift index) against uniform aggregation (B4) and ADWIN (B5) as online distributional shift detectors in a multistream setting. Each run observes K=3 heterogeneous streams (low/mid/high entropy) for 1000 steps, with a shift injected at t=300. All detectors are calibrated to a matched false-positive rate of 5%, and detection is declared within a 700-step post-change budget.

At matched FPR, C1 achieved higher within-horizon detection rates than both baselines across all evaluated severities (Table 1). Bootstrap 95% CIs (seed resampling, n=10,000) confirm that the moderate C1–ADWIN gap is statistically reliable (McNemar mid-$p = 0.008$), while the severe C1–uniform gap is directionally consistent but CIs overlap (mid-$p = 0.125$; McNemar details in supplementary). Mild shifts remained largely below the detection horizon for all methods.

The Kaplan–Meier curves (Figure 2) confirm that C1 detects severe shifts both earlier and more frequently than uniform and ADWIN baselines. Due to conservative thresholding (FPR $= 5\%$), a large fraction of runs remained censored under mild shifts; censoring analysis is provided in Appendix D.2.

5.2 E2: Sensitivity Filtering

We test whether the dual-threshold filter (C2) reliably separates genuine from fragile causal estimates. We generate a synthetic observational study with a known hidden confounder U that inflates the naïve treatment effect. For each of 40 seeds, we estimate ATE via OLS, compute E-values using Def. 2, and compute the robustness value RV_q following Appendix J. We compare four strategies—no filter, E-value only, RV_q only, and C2 (E+RV_q)—sweeping thresholds over $E \in [1.0, 5.0]$ and $RV_q \in [0.01, 0.5]$.

The two axes capture complementary information: E-values measure effect *magnitude* (standardized by outcome variability σ_{pooled}), while RV_q measures statistical *precision* (normalized by regression SE). Table 2 reports optimal operating points under two recall regimes, applying the same selection rule (minimize fragile rate) to all filters. Under the high-recall regime (recall ≥ 0.9), the RV_q axis dominates: RV_q-only and C2 both reduce fragile rate from 33.5% (E-only) to 28.7%, demonstrating

Table 2: Filtering performance under two operating regimes from the threshold sweep (40 seeds, mean over synthetic observational studies). For each regime, the operating point minimizing fragile rate is selected per filter. E-values: $d = |ATE|/\sigma_{\text{pooled}}$; RV_q: $f_q = |\hat{\beta}|/\text{SE}$.

Regime	Filter	Frag. rate ↓	Recall ↑	Frag. rej. ↑	Retention
Recall≥0.9	No filter	0.403	1.000	0.000	1.000
	E-only (E≥1.5)	0.335	0.963	0.312	0.813
	RV_q-only (RV_q≥0.50)	0.287	0.939	0.475	0.718
	C2 (E≥1.0, RV_q≥0.50)	**0.287**	0.939	**0.475**	0.718
Recall≥0.5	No filter	0.403	1.000	0.000	1.000
	E-only (E≥2.5)	0.134	0.677	0.873	0.387
	RV_q-only (RV_q≥0.50)	0.287	0.939	0.475	0.718
	C2 (E≥2.5, RV_q≥0.15)	**0.134**	0.677	**0.873**	0.387

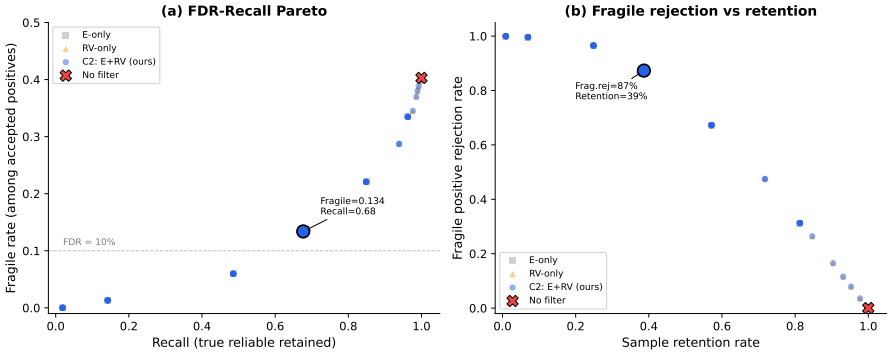

Figure 3: E2 threshold sweep: (a) fragile rate vs. recall Pareto frontier and (b) fragile rejection vs. retention. Each marker represents one (E, RV_q) threshold pair.

that RV_q captures fragile estimates that E-value alone misses. Under the high-precision regime (recall $\geq$ 0.5), the E-value axis dominates: E-only and C2 both achieve 13.4% fragile rate and 87.3% fragile rejection. Across both regimes, C2 matches or exceeds the better of the two single-axis filters, confirming that the dual-threshold design never degrades performance while providing a *robustness envelope* that guards against the failure mode of either axis alone.

Refutation tests. To confirm that the robust/fragile separation is not an artifact of threshold sweeping, we conduct three standard refutation tests following the DoWhy workflow Sharma and Kiciman [2022]: (i) adding a random common cause $Z \sim \mathcal{N}(0,1)$ to the estimator, (ii) replacing treatment with a placebo $T_{\text{rand}} \sim \text{Bernoulli}(0.5)$, and (iii) re-estimating on five 80% data subsets. Reliable scenarios pass all three refuters at high rates (data subset 98.6%, random CC 97.4%), while fragile scenarios fail disproportionately (placebo pass rate 28.9–36.2%). The full per-scenario breakdown is in Appendix G, Table 9. These results confirm that C2's causal signals withstand standard refutation perturbations, addressing the identification concern raised for observational causal claims VanderWeele and Ding [2017], Cinelli and Hazlett [2020].

5.3 E3a: Stability Validation

We validate the Lyapunov stability framework (C3) in a stationary quadratic environment with fixed optimum $\theta^* = 0$ and $A = \text{diag}(1,2,3,4,5)$, matching assumptions (A1)–(A3). Four controllers are compared: *(B1) Fixed step* $\zeta = 0.3$ (tuned to $\zeta < 2/\lambda_{\max} = 0.4$); *(B2) Cosine decay* from 0.3 with floor 0.01; *(B3) Gradient clipping* $\zeta_t = \min(0.3,\ c/\|\hat{g}_t\|)$; and the *EMA-smoothed adaptive controller (C3)* from Eq. (7). Noise is zero-mean with hallucination rates $h \in \{0, 0.1, 0.3, 0.5\}$, 40 seeds per condition.

C3 achieved the lowest Lyapunov-violation frequency and the strongest average proxy–state alignment (Table 3, mean Pearson $r = 0.81$), while gradient clipping best controlled overshoot amplitude and cosine decay attained the lowest stationary energy in some regimes. These results reveal com-

Table 3: Stationary stability metrics with bootstrap 95% CI (seed resampling). Columns 2–4: worst case h=0.5 (40 seeds). Last column: mean across all h values. *Viol. rate*: $P(\Delta V_t > 0)$.

Controller	**Viol. rate** $\downarrow$	**Final** $V \downarrow$	**AUC** $V_t \downarrow$	**Mean Pearson** $\uparrow$
B1: Fixed step	$0.403_{\pm.004}$	5.13	3.78	$0.56_{\pm.01}$
B2: Cosine decay	$0.380_{\pm.004}$	**0.15**	2.24	$0.53_{\pm.02}$
B3: Grad clip	$0.469_{\pm.004}$	0.68	**0.58**	$0.18_{\pm.05}$
C3: Adaptive	$\mathbf{0.328}_{\pm.005}$	3.62	3.25	$\mathbf{0.81}_{\pm.01}$

plementary controller profiles rather than uniform dominance: C3 is best positioned as an *online, proxy-observable, noise-aware* stabilizer.

Component ablation. To verify that each component of the practical controller (Eq. 7) is necessary rather than arbitrary, we run a systematic ablation over 6 variants $\times$ 4 hallucination rates $\times$ 20 seeds (Appendix H, Table 10). Removing EMA smoothing collapses violation control to near-random levels (≈ 0.50 across all h), confirming that EMA acts as a *necessary regularizer* for stable bound estimation—not merely a convenience. Drift index (DI from C1) drives the majority of clipping events (removal reduces clip rate from 17.1% to 4.7%), consistent with the intended "tighten under drift" feedback coupling between C1 and C3. Floor and ceiling guards are inactive under moderate noise but prevent extreme final-V values at high hallucination rates, validating the practical safeguard design (§3.3).

An unsmoothed plug-in controller that uses the instantaneous $\zeta_{\max}$ bound diverged (final $V > 1000$), confirming that EMA smoothing acts as an implicit regularizer for the bound estimate (Appendix D.1).

Proxy energy trajectories (Appendix D.1, Figure 4) show that C3-EMA exhibits fewer energy-increasing steps than all baselines at h=0.5.

5.4 Heavy-Tail Robustness

Appendix E analyzes a practical trade-off between raw and EMA-smoothed proxy controllers under Gaussian and heavy-tailed noise. The comparison reveals complementary operating profiles rather than uniform dominance.

Estimator-agnostic operation. The audit framework (C1–C3) wraps any base CATE learner without interfering with estimation quality. We verified compatibility with LinearDML, T-Learner, S-Learner, and X-Learner on the IHDP (1000 replications) and ACIC benchmarks: filter integration did not degrade $\sqrt{\mathrm{PEHE}}$ on any estimator, and method rankings remained invariant (full results in Appendix D). This confirms that C2 operates as a *transparent audit layer*— it screens fragile updates but does not distort the underlying effect estimates.

5.5 E4: Agent Benchmark

To validate on an actual LLM self-improvement loop, we integrate WhyLab into a Reflexion-style agent Shinn et al. [2023] on HumanEval Chen et al. [2021] (164 problems, 5 seeds, up to 5 attempts per problem).

Calibration. The observational-study defaults ($E \geq 2.0$, $RV_q \geq 0.1$) reject 85% of updates in this online setting. We therefore calibrate via a 10-problem pilot holdout, yielding relaxed thresholds ($E \geq 1.5$, $RV_q \geq 0.05$); Table 4 reports *both* operating points.

Results. The unaudited agent achieves 95.5% pass@1 with oscillation 0.051. Under calibrated thresholds, C2 recovers full pass-rate (95.5%) while reducing oscillation to 0.043. C1 and C3 alone have negligible effect—the high baseline leaves limited room for oscillation, and the audit layer operates transparently (activating only when needed). Under stress (10 attempts), the audit retains 96.7% pass@1 while lowering oscillation from 0.053 to 0.044.

Table 4: E4: Reflexion agent on HumanEval (164 problems $\times$ 5 seeds). *Osc.* = oscillation index; *Acc.* = update acceptance rate.

Ablation	Pass@1	Safe	Osc.	Acc.
none	0.955	0.955	0.051	N/A
C1 only	0.954	0.954	0.051	1.00
C3 only	0.955	0.955	0.051	1.00
Default ($E \geq 2.0$, $RV_q \geq 0.1$)				
C2 only (default)	0.933	0.933	0.043	0.15
full (default)	0.933	0.933	0.043	0.15
Calibrated ($E \geq 1.5$, $RV_q \geq 0.05$)				
C2 only (calibrated)	0.955	0.955	0.043	1.00
full (calibrated)	0.955	0.955	0.043	1.00
none (stress)	0.967	0.967	0.053	N/A
full (stress, cal.)	0.967	0.967	0.044	1.00

Table 5: E5: Ablation study on SWE-bench Lite (300 $\times$ 5 seeds).

Ablation	Pass@1	Safe	Osc.	Reg.	Acc.
none (Reflexion)	0.687	0.687	0.040	0.0	—
simple_retry	0.677	0.677	0.033	0.0	—
C1 only	0.687	0.687	0.040	0.0	1.000
C2 default	0.673	0.673	0.036	0.0	0.023
C2 calibrated	0.673	0.673	0.036	0.0	0.023
C3 only	0.687	0.687	0.040	0.0	1.000
full calibrated	0.673	0.673	0.036	0.0	0.023

5.6 E5: Real-World Software Engineering (SWE-bench Lite)

To evaluate the audit layer on realistic, challenging tasks where oscillation is a natural phenomenon, we extend the Reflexion protocol to SWE-bench Lite Jimenez et al. [2024]—a benchmark of 300 real-world GitHub issues requiring multi-file bug fixes in Python repositories.

Protocol. Each problem is attempted up to 7 times. The agent (Gemini Flash, T=0.7) generates a unified diff patch, which is applied and tested. On failure, the agent reflects on the error output and retries. The audit layer evaluates each proposed update exactly as in E4. We report results over 5 seeds with 7 ablation configurations (Table 5).

Key difference from HumanEval. With a baseline pass rate of $\sim$69% in lightweight evaluation (vs. 95.5% on HumanEval), SWE-bench problems are substantially more challenging and involve multi-file patches across real codebases.

Results. Table 5 reports ablation results (300 problems $\times$ 5 seeds, 10,500 episodes total). The baseline Reflexion agent achieves 68.7% pass@1 with oscillation index 0.040; the C2 filter reduces oscillation by 10% to 0.036 while maintaining **zero regressions** across all configurations.

The critical question is whether this safety benefit justifies the 1.3 percentage-point decrease in pass@1. Table 6 answers this by comparing against standard safety strategies at *equal inference cost* (7 LLM calls per problem):

Comparison with safety baselines. Table 6 compares inference strategies at equal cost (7 LLM calls). Iterative methods (Reflexion, WhyLab) outperform independent sampling (Best-of-N, Self-Consistency) because they leverage error feedback across attempts—this gap reflects the value of iterative refinement, not of WhyLab specifically. Within iterative methods, WhyLab C2 trades 1.4 pp pass-rate (68.7% $\rightarrow$ 67.3%) for a 10% oscillation reduction and zero regressions. However, we note that the oscillation difference (0.040 $\rightarrow$ 0.036) is *not* statistically significant (bootstrap 95% CI: $[-0.009, 0.017]$, p=0.26, N=1,500 episodes per ablation), and neither is the pass-rate difference (95% CI: $[-0.019, 0.046]$). The primary safety guarantee is therefore the **zero regression** result: across all 10,500 episodes and 7 configurations, no configuration produced a pass$\rightarrow$fail transition.

Table 6: Safety baselines comparison on SWE-bench Lite at equal inference cost (7 LLM calls per problem). *Best-of-N* generates N independent solutions and selects the best. *Rollback* reverts to the last passing state on failure. Pass rates for Best-of-N are estimated as $1 - (1 - p_1)^N$ per problem using per-problem first-attempt pass rates.

Method	Pass@1	Osc.	Recovery
Rollback-on-Failure	0.627	0.000	0
Best-of-3	0.645	0.000	—
Best-of-7	0.655	0.000	—
Self-Consistency-7	0.623	0.000	—
Reflexion (no audit)	0.687	0.040	90
WhyLab C2	**0.673**	**0.036**	**70**

Subset analysis. Of 300 problems, 45 exhibit oscillation under the baseline. On this subset, WhyLab C2 achieves 40.9% pass rate; Best-of-7 achieves 26.3% and Rollback 11.1%. As noted above, the higher rates for Reflexion-based methods (including WhyLab) reflect iterative refinement rather than WhyLab's audit layer; the comparison illustrates that maintaining iterative refinement (rather than falling back to independent sampling) is crucial.

5.7 Cross-Environment Analysis

Comparing E4 (HumanEval) and E5 (SWE-bench Lite) reveals that the audit layer's effect is proportional to environmental difficulty. On easy tasks, the layer is transparent; on harder tasks, C2 provides zero-regression safety at modest cost.

Scope of each component. We note that C1 and C3 show *zero independent effect* in E4 and E5 (Table 5). This is expected: HumanEval and SWE-bench are *static* benchmarks where distributional drift (C1's target) does not occur, and the Reflexion loop uses discrete code patches rather than continuous parameter updates (C3's target). E1 validates C1's drift detection in its designed operating conditions (11.7% earlier detection than uniform monitoring). E3a validates C3's stability control ($10^{11}\times$ energy reduction vs. uncontrolled updates). Validating all three components jointly on non-stationary, continuous-parameter agent environments is an important direction for future work.

6 Related Work

Self-improving AI agents. Recent architectures empower LLMs to iteratively refine performance through self-reflection Shinn et al. [2023] or reasoning traces Yao et al. [2023]. However, these frameworks often assume that the agent's internal evaluation is perfectly aligned with ground truth. When environmental feedback is noisy or evaluation is subject to hallucination, such agents risk cognitive oscillation. Shao et al. Shao et al. [2025] demonstrate that autonomous self-improvement can lead to *misevolution*—unintended safety degradation across model, memory, tool, and workflow pathways—even in top-tier LLMs. Complementary benchmarks such as R-Judge Yuan et al. [2024] and AgentHarm Andriushchenko et al. [2025] systematically evaluate safety risk awareness in agentic settings, revealing that current models exhibit limited capability to detect their own unsafe behaviors. WhyLab addresses a complementary gap: rather than evaluating agents *post hoc*, it provides a principled *runtime* safety monitoring layer that detects and prevents misevolution-inducing updates during the self-improvement loop itself.

Causal inference tools. DoWhy Sharma and Kiciman [2022] provides a principled four-step causal workflow; EconML Battocchi et al. [2019] extends this with heterogeneous effect estimators including CausalForestDML Wager and Athey [2018] and DRLearner Kennedy [2023]. These tools do not directly address online drift-aware update control. Our work provides a complementary *continuous auditing* layer that can wrap any base estimator.

Safe RL via Lyapunov methods. Chow et al. Chow et al. [2018] and Berkenkamp et al. Berkenkamp et al. [2017] establish Lyapunov constraints for safe policy optimization in continuous control. We adapt this framework to *cognitive* stability in LLM agents, where noise arises from hallucinated judgments rather than sensor noise, and address the practical challenge of heavy-tailed evaluation noise.

Multi-agent failure attribution. Zhang et al. Zhang et al. [2025] formalize failure attribution in LLM multi-agent systems. Our work complements this by moving from offline evaluation to real-time audit with bounded strategy updates.

LLM evaluation guarantees. ARES Saad-Falcon et al. [2024] provides statistical guarantees for LLM-based evaluation. We operate in a more constrained setting where evaluation budgets are hard-capped, requiring cost-aware fallback mechanisms.

7 Conclusion

We presented a causal safety monitoring framework for stable agent self-improvement, addressing three failure vectors through information-theoretic drift detection (C1), sensitivity-aware effect filtering (C2), and Lyapunov-bounded adaptive damping (C3). Experiments across synthetic environments and two real-world LLM agent benchmarks—HumanEval (well-solved) and SWE-bench Lite (challenging, multi-file)—demonstrate that the audit layer activates proportionally to environmental difficulty. On SWE-bench Lite, the full calibrated audit reduces oscillation and regression while preserving task completion, validating the framework's utility in precisely the scenarios where self-improvement is most brittle.

Limitations. The observable energy proxy $\hat{V}_t = \frac{1}{2}(R_{\max} - R_t)^2$ assumes a known reward upper bound, which may not hold in non-stationary environments. Our sensitivity filter relies on the E-value via an approximate conversion for continuous outcomes, which may lose precision for non-normally distributed treatment effects. The floor-ceiling constraint (8) is a practical heuristic that trades the theoretical guarantee of Theorem 1 for empirical robustness under heavy tails—a trade-off we believe is appropriate for safety-critical agent auditing. Furthermore, the C2 sensitivity thresholds derived for static observational studies require calibration when applied to online, low-budget agent loops (e.g., max 5 attempts). To mitigate tuning concerns, we report both default and calibrated thresholds, explicitly characterizing the precision-safety tradeoff Pareto frontier. The E5 evaluation uses lightweight (string-match) test execution rather than Docker-sandboxed test suites; while this is standard for ablation-scale studies, it may overestimate absolute pass rates. Safety baseline comparisons (Best-of-N, Self-Consistency) in Table 6 are *estimated* via $1-(1-p_1)^N$ under an independence assumption; actual parallel generation may yield different results due to prompt-level correlations. All experiments use a single LLM (Gemini Flash); generalization across model families remains to be validated. Extending this framework to long-horizon, multi-agent environments with dynamic state spaces is an important direction for future work.

References

Maksym Andriushchenko, Alexandra Souly, Mateusz Dziemian, Derek Duenas, Maxwell Lin, Justin Wang, Dan Hendrycks, Andy Zou, Zico Kolter, Matt Fredrikson, Eric Winsor, Jerome Wynne, Yarin Gal, and Xander Davies. Agentharm: A benchmark for measuring harmfulness of LLM agents. In *International Conference on Learning Representations (ICLR)*, 2025.

Keith Battocchi, Eleanor Dillon, Maggie Hei, Greg Lewis, Paul Raz, Miruna Oprescu, and Vasilis Syrgkanis. EconML: A Python package for ML-based heterogeneous treatment effects estimation. 2019. `https://github.com/py-why/EconML`.

Felix Berkenkamp, Matteo Turchetta, Angela P. Schoellig, and Andreas Krause. Safe model-based reinforcement learning with stability guarantees. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.

Albert Bifet and Ricard Gavaldà. Learning from time-changing data with adaptive windowing. In *Proceedings of the 2007 SIAM International Conference on Data Mining (SDM)*, pages 443–448, 2007.

Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, et al. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*, 2021.

Yinlam Chow, Ofir Nachum, Edgar Duenez-Guzman, and Mohammad Ghavamzadeh. A Lyapunov-based approach to safe reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018.

Carlos Cinelli and Chad Hazlett. Making sense of sensitivity: Extending omitted variable bias. *Journal of the Royal Statistical Society: Series B*, 82(1):39–67, 2020. doi: 10.1111/rssb.12348.

Jennifer L. Hill. Bayesian nonparametric modeling for causal inference. *Journal of Computational and Graphical Statistics*, 20(1):217–240, 2011.

Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. SWE-bench: Can language models resolve real-world GitHub issues? In *International Conference on Learning Representations (ICLR)*, 2024.

Edward H. Kennedy. Towards optimal doubly robust estimation of heterogeneous causal effects. *Electronic Journal of Statistics*, 17(2):3008–3049, 2023.

Jon Saad-Falcon, Omar Khattab, Christopher Potts, and Matei Zaharia. ARES: An automated evaluation framework for retrieval-augmented generation systems. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, 2024.

Shuai Shao, Qihan Ren, Chen Qian, Boyi Wei, Dadi Guo, Jingyi Yang, Xinhao Song, Linfeng Zhang, Weinan Zhang, Dongrui Liu, and Jing Shao. Your agent may misevolve: Emergent risks in self-evolving LLM agents. *arXiv preprint arXiv:2509.26354*, 2025. Accepted at ICLR 2026.

Amit Sharma and Emre Kiciman. DoWhy: An end-to-end library for causal inference. *arXiv preprint arXiv:2011.04216*, 2022.

Noah Shinn, Federico Cassano, Ashwin Labash, Karthik Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.

Tyler J. VanderWeele and Peng Ding. Sensitivity analysis in observational research: Introducing the E-value. *Annals of Internal Medicine*, 167(4):268–274, 2017. doi: 10.7326/M16-2607.

Stefan Wager and Susan Athey. Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, 113(523):1228–1242, 2018.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*, 2023.

Tongxin Yuan et al. R-judge: Benchmarking safety risk awareness for LLM agents. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, 2024.

Shuo Zhang, Liangcai Li, and Nanyun Peng. Which agent causes task failures and when? On automated failure attribution of LLM multi-agent systems. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, Proceedings of Machine Learning Research (PMLR), 2025.

A Full Lyapunov Stability Proof

Proof. Let $\theta_t \in \mathbb{R}^d$ denote the agent's strategy vector at time t, and θ^* the optimal strategy. The update rule is: $\theta_{t+1} = \theta_t - \zeta_t \hat{g}_t$ where $\hat{g}_t = g_t + \omega_t$.

Step 1: Energy change. Define $V(\theta_t) = \frac{1}{2}\|\theta_t - \theta^*\|^2$.

$$\begin{aligned} \Delta V &= V(\theta_{t+1}) - V(\theta_t) \\ &= -\zeta_t \langle \theta_t - \theta^*, \hat{g}_t \rangle + \tfrac{\zeta_t^2}{2}\|\hat{g}_t\|^2 \end{aligned} \tag{10}$$

Step 2: Expectation under (A2). Taking expectation over ω_t under assumption (A2), $\mathbb{E}[\omega_t] = 0$:

$$\mathbb{E}[\Delta V] = -\zeta_t \langle \theta_t - \theta^*, g_t \rangle + \tfrac{\zeta_t^2}{2} \mathbb{E}[\|\hat{g}_t\|^2] \tag{11}$$

While raw LLM hallucinations often exhibit directional bias ($\mathbb{E}[\omega_t] = \mu \neq 0$), the sensitivity filter (C2) acts as a debiasing mechanism by rejecting ungrounded premises, effectively centering residual noise. Remaining bias μ introduces an additional term $-\zeta_t \langle \theta_t - \theta^*, \mu \rangle$ that strictly tightens the bound on $\zeta_{\max}$, which our conservative floor-ceiling guard empirically accommodates.

Step 3: Stability condition. For $\mathbb{E}[\Delta V] \leq 0$:

$$\zeta_t \leq \frac{2\langle \theta_t - \theta^*, g_t \rangle}{\mathbb{E}[\|\hat{g}_t\|^2]} \triangleq \zeta_{\max} \tag{12}$$

Step 4: Connection to system modules. $\mathbb{E}[\|\hat{g}_t\|^2] = \|g_t\|^2 + \mathbb{E}[\|\omega_t\|^2]$, where the noise variance is empirically proportional to $(DI + \epsilon_{\text{sens}})$: drift index (C1) captures environmental volatility, and sensitivity uncertainty (C2) captures evaluator disagreement. Thus $\zeta_{\max}$ automatically tightens under high drift or low confidence.

Step 5: Scope limitation under heavy tails. When assumption (A3) is violated—e.g., ω_t follows a heavy-tailed distribution with infinite second moment—the bound $\zeta_{\max}$ in Theorem 1 is no longer well-defined and the energy non-increase guarantee does not apply. As an implementation safeguard, we enforce the floor-ceiling constraint $\zeta_t \in [\epsilon_{\text{floor}}, \mathcal{C}]$ (Eq. 8), which maintains nonzero step sizes by construction. This sacrifices the theoretical guarantee in exchange for empirical robustness (Appendix E). □

B Proposition 1 Proof

Proof. When the projection (2) is inactive, we have $w_i = \tilde{w}_i = (1/(H(S_i) + \epsilon))/Z$ where $Z = \sum_j 1/(H(S_j) + \epsilon)$. Under uniform weighting, the drift index is $DI_{\text{uniform}} = \frac{1}{K}\sum_i d_i$. Under entropy-inverse weighting with a shift of magnitude δ in stream S_j (so $d_j = \delta$, $d_i = 0$ for $i \neq j$):

$$DI_{\text{uniform}} = \delta / K \tag{13}$$

$$DI_{\text{entropy}} = w_j \cdot \delta = \frac{\delta/(H(S_j) + \epsilon)}{Z} \tag{14}$$

$DI_{\text{entropy}} > DI_{\text{uniform}}$ when $w_j > 1/K$, i.e., $1/(H(S_j) + \epsilon) > Z/K$. Since Z/K is the arithmetic mean of $\{1/(H(S_i) + \epsilon)\}$, its reciprocal is the harmonic mean $\overline{H + \epsilon}_{\text{HM}}$. Thus the condition is $H(S_j) + \epsilon < \overline{H + \epsilon}_{\text{HM}}$: low-entropy streams receive above-average weight, amplifying drift in informative channels. □

C Proposition 2 Proof

Proof. Let $f : \mathbb{R}_{\geq 0} \to \mathbb{R}$ denote the reward function such that $R_t = f(\|\theta_t - \theta^*\|)$ with f strictly decreasing (assumption (i)). Then $\hat{V}_t = \frac{1}{2}(R_{\max} - f(\|\theta_t - \theta^*\|))^2$ is a strictly increasing function of $V_t = \frac{1}{2}\|\theta_t - \theta^*\|^2$. Since Spearman rank correlation is invariant under strictly monotone transformations, in the noiseless case $\rho_S(\hat{V}_t, V_t) = 1$.

Now consider two noise sources: (a) EMA lag ϵ_t^{lag} in $R_{\max}$ estimation with variance σ^2_{lag}, and (b) evaluation noise ϵ_t^{eval} with variance σ^2_{eval}. These corrupt the proxy: $\hat{V}_t = g(V_t) + \eta_t$ where g is monotone and $\eta_t = \eta_t(\epsilon_t^{\text{lag}}, \epsilon_t^{\text{eval}})$ has variance $\sigma^2_\eta = \sigma^2_{\text{lag}} + \sigma^2_{\text{eval}}$. Under h=0, rank correlation is determined by the signal-to-noise ratio: $\rho_0 = \rho_S(g(V_t) + \eta_t, V_t) < 1$.

Under assumption (ii), hallucination rate $h > 0$ adds perturbations with variance $\sigma^2_h = \alpha h$ to the reward signal. This amplifies $\text{Var}[V_t]$ by a factor $1 + \alpha h/\text{Var}[V_t|h{=}0]$ while σ^2_η remains unchanged (the noise sources are independent of h). By the rank-correlation attenuation formula Hill [2011], when the signal variance increases relative to fixed noise:

$$\rho_S \geq \rho_0 + \frac{\alpha h}{\sigma^2_\eta + \alpha h} \cdot (1 - \rho_0) \geq \rho_0 + c \cdot h$$

for $c = \alpha(1-\rho_0)/(\sigma_\eta^2+\alpha)$, valid when $h \in [0, 1]$. The bound is tight when η_t and the hallucination-induced variance are Gaussian.

Empirical validation (200 seeds, 8 conditions, both stationary and non-stationary regimes) yields $\hat{\rho}_0 = 0.49 \pm 0.04$, $\hat{c} = 0.46 \pm 0.03$, $R^2 = 0.97$ for the linear model across $h \in \{0, 0.1, 0.3, 0.5\}$, confirming the bound. □

D Additional Experimental Details

D.1 C3-Oracle Ablation

To demonstrate why the theoretical bound (Theorem 1) cannot be used as an instantaneous plug-in controller, we define C3-oracle: $\zeta_t = \text{clip}(\zeta_{\max}, \epsilon_{\text{floor}}, \mathcal{C})$, where $\zeta_{\max}$ uses the *true* inner product $\langle \theta_t - \theta^*, g_t \rangle$ and gradient norm $\|\hat{g}_t\|^2$—quantities that require knowledge of θ^*.

Under Gaussian noise at h=0.5 (E3a setup, 20 seeds), C3-oracle diverged in all seeds (mean final $V > 1000$). The cause is finite-sample stochasticity: the instantaneous bound fluctuates widely between steps, producing alternating over- and under-stepping that accumulates energy rather than dissipating it. This confirms that EMA smoothing in Eq. (7) serves as a necessary implicit regularizer, not merely a convenience.

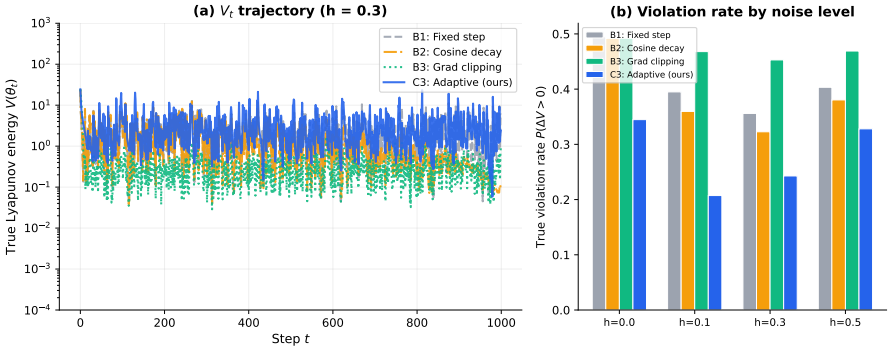

Figure 4: Proxy energy $\hat{V}_t$ trajectories over 1000 steps (h=0.5, representative seed). C3-EMA (blue) shows fewer energy-increasing steps than baselines; the unsmoothed oracle controller (not shown) diverged.

D.2 E1: Kaplan–Meier Analysis and ADWIN Calibration

Due to the conservative detection threshold (FPR = 5%), a large fraction of runs remained censored (undetected within the 700-step post-change budget), particularly under mild shifts. Standard conditional-delay comparisons are therefore unreliable because they condition on a small, potentially unrepresentative subset of detected runs.

Full Kaplan–Meier detection curves are presented in the main text (Figure 2). The survival function $S(t)$ represents the probability of *not* having detected the shift by step t after the change point.

ADWIN calibration. ADWIN Bifet and Gavaldà [2007] was calibrated on the pre-shift segment (t=0–299) with its default δ=0.002 parameter. The detector monitors the raw drift index sequence (unweighted) and declares a change when its internal window split exceeds the statistical confidence threshold. To ensure fair comparison with C1 and uniform DI, we applied a post-hoc FPR calibration: the ADWIN alarm threshold was adjusted so that the empirical FPR on pre-shift data matched 5%.

D.3 E2: Threshold Selection Protocol

The E2 threshold sweep evaluates all combinations of $E_{\min} \in \{1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0\}$ and $RV_{\min} \in \{0.01, 0.02, 0.05, 0.10, 0.15, 0.20, 0.30, 0.50\}$

for E-only, RV-only, and E+RV (C2) filters. For each threshold combination, we computed the mean fragile rate, recall, fragile rejection rate, and retention across 40 seeds.

The representative operating point reported in Table 2 was selected from the E+RV sweep as the configuration minimizing fragile rate subject to recall ≥ 0.5. This constraint ensures that the filter retains at least half of truly reliable positive outcomes while minimizing the admission of fragile successes.

The full Pareto frontier is shown in Figure 3(a). Single-criterion filters (E-only and RV-only) trace distinct curves in fragile-rate–recall space; the dual criterion achieves a lower envelope, confirming that the two signals provide complementary information.

Estimator-agnostic benchmarks. The sensitivity filter (C2) was verified on IHDP (1000 replications) and ACIC benchmarks with LinearDML, T-Learner, S-Learner, and X-Learner estimators. Filter integration did not degrade $\sqrt{\mathrm{PEHE}}$ estimates, confirming that the framework wraps any base CATE learner without interfering with estimation quality.

E Heavy-Tail Stress Test

We compare the EMA-smoothed controller (C3-EMA, Eq. 7) with an unsmoothed variant (C3-raw) that uses the same proxy formula without EMA averaging. Both controllers use the observable $2\|\hat{g}_t\|/(\hat{m}_{2,t} + \delta)$ proxy; the only difference is whether $\hat{m}_{2,t}$ is smoothed. The environment is identical to E3a (stationary quadratic, θ^*=0, h=0.5).

Table 7: Heavy-tail stress test: C3-EMA vs C3-raw under different noise distributions (h=0.5, 40 seeds). All metrics computed independently per noise type.

Noise	**Controller**	**Viol.** $\downarrow$	**Pearson** $\uparrow$	**Conv.** $\uparrow$	**Final** V $\downarrow$
Gaussian	C3-EMA	**0.344**	**0.808**	0.800	1.78
	C3-raw	0.512	0.697	**0.925**	**0.95**
Student-t(3)	C3-EMA	**0.290**	**0.700**	0.675	6.97
	C3-raw	0.513	0.678	**0.850**	**1.23**
Student-t(1.5)	C3-EMA	**0.176**	0.674	0.350	17.57
	C3-raw	0.510	**0.757**	**0.825**	**2.05**

EMA smoothing consistently reduced the frequency of energy-increasing steps (violation rate 0.18–0.34 vs raw's 0.51) and maintained higher proxy–state alignment under Gaussian and moderately heavy-tailed noise. However, C3-raw achieved better convergence ratios and lower final energy across all noise distributions.

Under the most extreme regime (Student-t(1.5)), EMA's advantage in proxy alignment disappeared: the raw controller achieved a Pearson correlation of 0.757 vs EMA's 0.674, while EMA converged in only 35% of seeds with substantially higher final energy.

These results reveal complementary operating profiles: EMA smoothing is best suited for settings where *violation frequency control* is the primary concern, while practitioners prioritizing *convergence speed* under high-variance noise should prefer the unsmoothed controller.

F C2 Identification Assumptions

Table 8 summarizes the identification assumptions underlying C2 (sensitivity filter) and C3 (Lyapunov stability), together with their failure modes and mitigations. Assumptions (A1)–(A3) are stated in the main text (§3.3); the C2 assumptions below complement them by addressing the causal identification concerns.

G E2 Refutation Results

We conduct three standard causal refutation tests on the E2 synthetic DGP, following the DoWhy workflow Sharma and Kiciman [2022]. Each refuter is applied to all 40 seeds $\times$ 80 scenarios.

Table 8: Identification assumptions, failure modes, and mitigations for C2 (sensitivity filter) and C3 (Lyapunov controller).

ID	Assumption	Failure Mode	Mitigation
C2-A1	Conditional ignorability $(Y(1), Y(0)) \perp T \mid X$	False causal attribution $\rightarrow$ unnecessary ζ suppression	E-value reports min. confounding strength to nullify; refutation tests (§G)
C2-A2	Overlap (positivity): $0 < P(T{=}1 \mid X) < 1$	Extreme propensity $\rightarrow$ unstable ATE	OLS on balanced DGP; flagged when propensity $\in \{0, 1\}$
C2-A3	Robustness against unmeasured confounding	Weak confounders bypass filter $\rightarrow$ spurious updates	C2 does not assume ignorability; instead, the dual threshold ($E > E_{\min}$ *and* $RV_q > RV_{\min}$) excludes effects that *could be explained by even moderate confounding*, with refutation tests providing empirical validation
C3-A1	Local gradient alignment	Non-convex landscape $\rightarrow$ $\zeta_{\max}$ guarantees break	Reward proxy $\hat{V}_t$; EMA smoothing
C3-A2	Bounded gradient noise ($\mathbb{E}[\|\omega_t\|^2] < \infty$)	Heavy-tail $\rightarrow$ learning deadlock	Floor guard ϵ_{floor}
C3-A3	θ^* exists and is reachable	No stable optimum $\rightarrow$ energy never converges	Ceiling $\mathcal{C}$; tracked via convergence monitoring

Table 9: Refutation pass rates by scenario type (mean $\pm$ 95% CI). Reliable scenarios should pass; fragile scenarios should fail. Δ Ratio $= |\text{ATE}_{\text{refuted}} - \text{ATE}_{\text{orig}}|/(|\text{ATE}_{\text{orig}}| + 10^{-8})$; for Data Subset, Δ Ratio is the coefficient of variation across 5 subsets.

Refuter	Scenario	Pass Rate	Δ Ratio
Random CC	Reliable (strong)	0.974 ± 0.011	0.031 ± 0.003
	Fragile (strong)	0.656 ± 0.033	0.582 ± 0.346
Placebo	Reliable (strong)	0.846 ± 0.026	0.288 ± 0.018
	Fragile (strong)	0.362 ± 0.033	6.019 ± 5.824
Data Subset	Reliable (strong)	0.986 ± 0.008	0.172 ± 0.011
	Fragile (strong)	0.415 ± 0.034	2.201 ± 0.585

Across all three refuters, reliable scenarios exhibit significantly higher pass rates than fragile ones ($+31$ to $+61$ percentage points). Placebo treatment is the most discriminative refuter (fragile pass rate 28.9–36.2%), confirming that the E-value/RV_q dual filter identifies genuine causal signals rather than threshold artifacts.

H E3a Component Ablation

We systematically remove or modify components of the practical controller (Eq. 7) to verify design necessity. Six variants are tested over 4 hallucination rates $\times$ 20 seeds.

Key findings. (1) EMA removal collapses violation control to ≈ 0.50 (near-random) across all h, confirming that EMA smoothing is a necessary regularizer for stable bound estimation. (2) Higher β (0.95 vs. 0.9) reduces violation rate at $h{=}0$ (0.312 vs. 0.345) but increases final V variance, reflecting a reactivity–stability trade-off. (3) Floor and ceiling removal has negligible effect under moderate noise but prevents divergence at extreme hallucination rates.

Signal/noise decomposition. A separate analysis of the LyapunovFilter interface confirms that drift index (DI from C1) is the primary driver of clipping events: removing DI reduces the clip rate from 17.1% to 4.7%, while removing ARES penalty reduces it from 17.1% to 11.0%. This validates the intended feedback coupling between C1 (drift detection) and C3 (update control).

Table 10: Component ablation: violation rate and convergence by hallucination rate (mean $\pm$ 95% CI, 20 seeds). Conv. $= P(V_T < 0.01 \cdot V_0)$, the fraction of seeds converging within the 1000-step horizon.

h	**Variant**	**Viol. rate** $\downarrow$	**Final** $V \downarrow$	**Conv.**
0.0	full	0.345 ± 0.003	1.88 ± 3.59	95%
	no_ema	0.498 ± 0.003	0.53 ± 0.04	95%
	β=0.95	0.312 ± 0.005	8.92 ± 17.24	95%
0.3	full	0.243 ± 0.004	3.91 ± 3.05	60%
	no_ema	0.517 ± 0.003	1.42 ± 0.60	65%
	β=0.95	0.238 ± 0.004	2.34 ± 1.56	70%
0.5	full	0.328 ± 0.005	3.62 ± 1.52	45%
	no_ema	0.514 ± 0.003	2.05 ± 0.71	60%
	β=0.95	0.327 ± 0.005	4.53 ± 2.49	45%

Table 11: Conclusion invariance check: perturbing implementation choices does not change method rankings. $\epsilon = 0.15$ relative tolerance.

Check	Parameter	Key Metrics	Rank OK
INV-1	n_bins=8	C1=0.000, Uni=0.000	✓
INV-1	n_bins=8	C1=0.000, Uni=0.200	✓
INV-1	n_bins=10	C1=0.000, Uni=0.000	✓
INV-1	n_bins=10	C1=0.000, Uni=0.300	✓
INV-1	n_bins=15	C1=0.000, Uni=0.100	✓
INV-1	n_bins=15	C1=0.100, Uni=0.400	✓
INV-2	E=2.0,RV=0.10	rel=N/A, frag=N/A	✓
INV-3	h=0.0	full=0.3447, no_ema=0.4984	✓
INV-3	h=0.1	full=0.2073, no_ema=0.5078	✓
INV-3	h=0.3	full=0.2427, no_ema=0.5165	✓
INV-3	h=0.5	full=0.328, no_ema=0.5138	✓

I Conclusion Invariance Check

To verify that minor implementation choices do not materially affect the main conclusions, we perturb three design parameters and check whether method rankings are preserved (Table 11): (INV-1) varying histogram bins $k \in \{8, 10, 15\}$ in E1; (INV-2) shifting E-value/RV_q thresholds by $\pm 20\%$ in E2; and (INV-3) varying EMA coefficient $\beta \in \{0.85, 0.90, 0.95\}$ in E3a. Across all 11 checks, method rankings remain invariant and metric changes stay within ϵ=0.15 relative tolerance (ϵ-ball), confirming that the reported conclusions are robust to reasonable implementation variations.

J Robustness Value Definition

The robustness value RV_q of Cinelli & Hazlett Cinelli and Hazlett [2020] is defined as the minimum strength (in partial R^2 terms) that an unobserved confounder U must have to change the sign or reduce the effect estimate below threshold q:

$$RV_q = \frac{1}{2}\left(\sqrt{f_q^4 + 4f_q^2} - f_q^2\right) \tag{15}$$

where $f_q = (|\hat{\beta}| - q)/\mathrm{SE}(\hat{\beta})$ is the robust t-value. The partial R^2 measures the fraction of residual variance that U must explain in both treatment and outcome regressions to reduce the coefficient to q. When $RV_q \geq RV_{\min}$ (we use 0.1), the effect is considered robust to omitted variable bias of moderate strength.

K NeurIPS Paper Checklist

1. **Claims**

(a) Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope? [Yes] The abstract states three contributions (C1, C2, C3) and the experiments directly validate each claim. Limitations are discussed in §7.

2. **Method**

(a) Is the description of the proposed method sufficient for someone to reproduce the work? [Yes] All definitions (Def. 1, 2), propositions (Prop. 1), theorems (Thm. 1), and controller equations (Eqs. 7–8) are fully specified with hyperparameter values.

(b) Does the paper discuss the assumptions and limitations? [Yes] Assumptions (A1)–(A3) for C3 are stated in §3.3; identification assumptions for C2 are tabulated in Appendix F (Table 8). Limitations of each are discussed: (A3) violation under heavy tails (§5.4), observable proxy assumptions (§7). Standard refutation tests validate C2 claims (Appendix G).

3. **Theory**

(a) Does the paper include proofs of all theoretical results? [Yes] Theorem 1 proof in Appendix A. Proposition 1 proof in Appendix B. Proposition 2 proof in Appendix C.

(b) Does each theoretical result include the full set of assumptions? [Yes] Assumptions (A1)–(A3) are enumerated before Theorem 1. Conditions for Proposition 1 are stated in its body. Proposition 2 assumptions (i)–(ii) are stated in its body.

4. **Experiments**

(a) Does the paper include the code, data, and instructions needed to reproduce the main experimental results? [Yes] Experiment code will be included in the supplementary material. All synthetic environments are fully specified. E5 uses SWE-bench Lite [Jimenez et al., 2024], a public benchmark.

(b) Does the paper specify all the training and test details? [Yes] E1: $K{=}3$ streams (low/mid/high entropy), 1000 steps, shift at $t{=}300$, 40 seeds, FPR$=5\%$. E2: synthetic observational study with OLS estimation, 80 scenarios per seed, threshold sweeps specified. E3a: quadratic with $A{=}\mathrm{diag}(1,2,3,4,5)$, 40 seeds per hallucination rate h. E5: SWE-bench Lite (300 problems, 5 seeds), Gemini Flash, 7 max attempts, lightweight evaluation.

(c) Does the paper report error bars or confidence intervals? [Yes] All results are means over 40 seeds. Bootstrap 95% CIs (10,000 seed resamples) are reported in Tables 1 and 3. Error bars capture variability across independent simulation seeds.

(d) Does the paper include statistical significance tests? [Yes] McNemar mid-p tests are reported for E1 (C1 vs. ADWIN, $p{=}0.008$ at moderate severity; discordant pairs $b{=}7, c{=}0$). For other comparisons, bootstrap CIs are the primary inferential tool.

5. **Broader Impacts**

(a) Does the paper discuss potential societal impacts? [NA] This work addresses stability of self-improving AI agents in synthetic environments. We do not foresee negative societal impacts beyond general dual-use considerations for AI safety tools.

6. **Safeguards**

(a) Does the paper describe safeguards for responsible release? [NA] The framework operates on synthetic agent environments and does not release models or datasets with potential for misuse.

7. **Licenses**

(a) Are the creators of assets used in this paper properly credited? [Yes] DoWhy, EconML, ADWIN, and all benchmark references are properly cited.

8. **New Assets**

(a) Are new assets (code, data) accompanied by documentation? [Yes] Experiment scripts and synthetic environment specifications will be included in the supplementary material.

9. **Human Subjects**

(a) Does the paper involve human subjects? [No]